

- Anaphase ---- Centromere divides
 Metaphase --- chromosomes migrate to opposite poles
 Prophase ---- Condensation of chromosomes

12. Which is wrong about the process of cross over?

- A. A chiasma is formed at the point of interchange of chromatid segments of homologous chromosomes
 B. It occurs at meiotic II
 C. Results genetic variability
 D. Random distribution of homologous chromosomes to the daughter cells
 E. None of the above

Instruction II: Matching Questions: Match items from column "B" to items in column "A"

Column "A"

Column "B"

- | | |
|---|------------------------|
| 1. Transverse movement of phospholipids F | A. Cholesterol |
| 2. Lateral movement of phospholipids E | B. Phosphatidylserine |
| 3. Preferential component of inner leaflet of membrane B | C. Phosphatidylcholine |
| 4. Sphingolipid component of myelin sheath D | D. Sphingomyelin |
| 5. Modulator membrane fluidity A | E. Fast movement |
| | F. Slow movement |

Instruction III. Fill in the blank questions

1. Galactose and mannose are formed by epimerization of glucose.
2. N-acetylglucuronic acid is a uronic acid formed by oxidation of C-6 of glucose into carboxyl group.
3. Arachidonic acid can be formed in our body from an essential fatty acid known as linoleic acid.
4. Glyceralphosphate, Glucosamine is an example of amino sugar.

3

- A. Affecting hydrogen bonding
 - B. Disrupting hydrophobic interactions
 - C. Forming ionic bonding
 - D. Protonating key amino acid residues
 - E. Breaking peptide bonds
63. Which one is not true about denaturation of DNA?
- ☒ A. Covalent bonds will break
 - B. Physical changes because of high temperature
 - C. Hydrogen bonding is disrupted
 - D. A-T base pair more resistant than G-C base pair
 - E. Ionic interactions dissociate
64. Which one is not correctly matched?
- A. Caudal/inferior ✓
 - B. Ventral/anterior ✓
 - C. Dorsal/posterior ✓
 - ☒ D. Cranial/superior ✓
 - ☒ E. None of the above
66. Wrong about thalidomide?
- A. Amelia is a teratologic effect of thalidomide
 - B. Influenced for the study of teratology that with causes birth defects
 - ☒ C. Used as antinuseant to pregnant women, resulting no birth defects
 - D. Affects the bones of the limbs
 - E. None of the above
67. Which one is not part of a uterine tube?
- A. Isthmus
 - B. Infundibulum
 - C. Ampulla
 - D. Cervix
 - E. Intrauterine part
68. Which of the following is not found in the pelvis?
- A. Testes
 - ☒ B. Prostate gland
 - C. Seminal vesicles
 - D. Ductus deferens
 - E. Ejaculatory duct
69. Which one is not found in the pelvis?
- A. Cervix
 - B. Vagina
 - C. ovary
 - D. Uterus
 - ☒ E. None of the above
70. Wrong about primordial germ cells (PGCs)?
- A. PGCs arise from the epiblast of the embryo ✓
 - B. During their migration to the gonads they increase their number with mitotic division
 - ☒ C. Enter into meiotic division during the fourth – fifth weeks of development
 - D. Formed during the first week of development ✓
 - E. Teratomas are tumors from abnormal development of PGCs ✓
71. Wrong match about the stages of mitosis?
- A. Metaphase--- double structured chromosomes become visible
 - B. Telophase --- chromosomes uncoil and elongate

- D. Muscle fiber - muscle - fascicle
E. None of the above
54. The amino acid which can serve as proton donor and acceptor in
A. Glycine
B. Proline
☒ C. Histidine
D. Isoleucine
E. Tryptophan
55. A semi-essential amino acid that is required especially during pregnancy is
A. Methionine
B. Valine
☒ C. Arginine
D. Leucine
E. Tyrosine
56. The most common type of amino acid that frequently present in β -turns is
A. lysine
☒ B. glycine
C. valine
D. isoleucine
E. tyrosine
57. The electrostatic bond of side chains to form specific shape of a protein is described by
A. Primary structure
B. Secondary structure
☒ C. Tertiary structure
D. Quaternary structure
58. Which one is true regarding the folding of large polypeptides from polysomes $\rightarrow$ complexes?
A. Folding starts in the ribosome
B. Its native folding is ATP independent
C. It is a random and spontaneous process
D. It has high tendency for aggregation
E. All of these
59. The role of molecular chaperones include the following except
A. They shield interactive surface of non-native polypeptide
B. Mostly expressed constitutively
C. Recognize hydrophilic residues
D. They can reverse intramolecular misfolding
E. Facilitate assembly/disassembly of multiprotein complexes
60. Chemical denaturing agents of proteins include
A. Ionizing radiation
B. Extreme pH
C. Heat
D. Vigorous shaking
☒ E. All of these
61. The level of protein structure disrupted during denaturation include the following except
☒ A. Primary structure
B. Secondary structure
C. Tertiary structure
D. Quaternary structure
E. Both C and D
62. Organic compounds usually affect the protein structure by

48. Which one is not true about Anatomical Position?
- A. Body erect ✓
 - B. Head, eyes and toes directed forward
 - C. Feet together
 - ☒ D. Arms at side with palms facing backward
 - E. None of the above
49. Flexion and extension movements are possible through:
- A. Sagittal axis
 - ☒ B. Horizontal axis
 - C. Vertical axis
 - D. Longitudinal axis
 - E. None of the above
50. Which one of the following movement is different from the others?
- A. Protrusion
 - ☒ B. Flexion
 - C. Extension
 - D. Abduction
 - E. Hyperextension
- ☒ 51. Which one of the following bone markings is not related to joints?
- A. Head
 - B. Facet
 - C. Condyle
 - ☒ D. Foramen
 - E. None of the above
52. All of the following statements are true about sesamoid bones except:
- A. Prevent excessive wear and tear
 - B. Helps tendons to glide over bony surfaces
 - C. Alter the direction of pull of the tendons
 - ☒ D. Decrease mechanical advantage in the joints of long bones
 - E. None of the above
53. Which one is a correct order in structural organization of skeletal muscle from cell to whole muscle?
- A. Fascicle - muscle fiber - muscle
 - ☒ B. Muscle fiber - fascicle - muscle
 - C. Muscle - muscle fiber - fascicle

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- ... important for 2^o active transport
- D. Creating membrane potential difference
- E. Avoid sodium gradient across the cell membrane
43. What happens to the rate of the transportation of a substance across the cell membrane in the facilitated diffusion after transport maximum is achieved at a given time
- ☒ A. Constant
- B. Increased
- C. Reduced
- D. Slightly reduced
- E. No transport maximum in the facilitated diffusion
44. Which of the following is true about body fluid
- A. More water removed by kidney during exercise
- B. No water is lost from the body without regulation
- C. Protein in the urine brings tissue edema
- D. The more the filtration coefficient is the less the filtration rate
- E. Capillary hydrostatic pressure gets high across the capillary
45. Which of the following is true about the composition of the intracellular and extra cellular fluid compartments
- ☒ A. Intracellular osmolarity is greater than the extracellular ↓
- B. More magnesium is outside the cell than the inside X
- C. More proteins found inside the cell than the outside X
- D. Potassium is the extracellular cation X
- E. More water is found outside the cell than the inside ✓
46. Which of the following electrolyte is found more in the blood vessels than in the tissue
- ☒ A. Calcium²⁺ cat
- B. Chloride⁻
- C. Sulphate⁻
- D. Bicarbonate ion⁻
- ☒ E. Phosphate⁻
47. Valves in the veins were discovered by:
- ☒ A. Hieronymus Fabricius
- B. William Harvey
- C. Andreas Vesalius
- D. Leonardo da Vinci
- E. Hippocrates

- A. The more the number of osmotically active particles is the less the osmotic pressure
- ☒ B. The number of the osmotically active particle for non ionized chemical is one ✓
- C. The osmotically active particle of urea is greater than the sodium chloride ✗
- D. Addition of calcium chloride into a solution reduces the osmotic pressure of the solution ✗
- E. The osmotic pressure of a solution with calcium chloride is less than the sodium chloride with the same concentration

Given a solution of 0.4% Na_2PO_4 with the following atomic weight: $\text{Na}^+ = 23$, $\text{P} = 31$ and $\text{O} = 16$.
Then answer the following questions

38. Osmolarity of the solution

- A. 0.1135 osm/L
B. 0.0850 osm/L
C. 0.0567 osm/L
D. 0.0650 osm/L
E. 0.0485 osm/L

39. Osmotic pressure of the solution

- A. 1550.7 mmHg
B. 1640.5 mmHg
C. 5475.1 mmHg
D. 1563.5 mmHg
E. None of the above

40. Which of the following is not the property of the simple diffusion through passive channels?

- A. Transport maximum ✓
B. Dissipate potential ✓
C. Leaky ion channels
D. Hydrophilic open field proteins
E. No energy needed ✓

41. Which of the following is true about the active transport across the cell membrane?

- A. The source of energy for 2^0 active transport is indirectly ions ✗
B. Calcium influx into sarcoplasmic reticulum is 2^0 active transport ✗
C. $\text{Na}^+ - \text{K}^+$ Exchange in the cardiac cell is direct active transport ✗
D. Vesicular bulk transport is passive transportation process ✗
E. Pinocytosis is non energy mediated transport ✗

42. Which of the following is not the function of the $\text{Na}^+ - \text{K}^+$ pumps

- A. Prevent intracellular edema
B. Create sodium diffusion gradient

- A. Are made from 13 protofilaments arranged circumferentially ✓
 B. Could increase in length, but not in width ✓
 C. Transport vesicles within their luminal space from one region to another region in the cell
 D. All of the above
 E. None of the above
32. Regarding the nucleolus:
 A. Is acidophilic
 B. Is the site of mitochondrial synthesis ✗
 C. Is membrane bound region within the nucleus
 D. Is made by regions of chromatins contributed by different chromosomes
 E. Is not visible under the light microscope
33. Organelles made by stalk of 3-10 discrete flattened and slightly curved bag-like channels or cisternae are:
 A. Mitochondria
 B. Smooth endoplasmic reticulum
 C. Peroxisomes
 D. Golgi bodies
 E. Microtubules
34. During a cell cycle,
 A. Duplication of centrioles begins and complete within the G₁ phase ✓
 B. The S phase is the longest and most variable phase ✗
 C. Daughter cells grow to the size of the parent cell during the G₁ phase ✗
 D. Synthesis of RNA and proteins occur mainly during the S phase ✗
 E. Replications of the chromosomes occur during G₂ phase ✗
35. Which of the following is true about transportation process across the cell membrane
 A. Conformational change is the property of permeases ✗
 B. Energy brings conformational changes on the leaky ion channels ✗
 C. ATP is the direct energy in the secondary active transport ✗
 D. All carrier mediated transport is active ✗
 E. Binding site on the channels are specific
36. All of the following increase the diffusion rate of a substances across the cell membrane except one
 A. The more the concentration gradient ✗
 B. The more the permeability of the membrane ✗
 C. The more the temperature of the substances ✗
 D. The more the thickness of the membrane
 E. The smaller the size of the substances ✗
37. Which of the following is true about the Van't Hoff's law

- A. Extensibility
- B. Excitability
- C. Contractility

- ☒ D. Elasticity
- E. All of the above

25. The muscle fiber arrangement that we find around the openings of the body is:

- A. Convergent
- B. Parallel
- ☒ C. Sphincter

- D. Pennate
- E. All of the above

26. All of the following cranial nerves carry sensory, motor and parasympathetic fibers except: *Craniosacral*

- A. Facial nerve
- B. Oculomotor nerve
- C. Glossopharyngeal nerve

- D. Vagus nerve
- E. None of the above

27. A membrane which covers body cavities confined to the internal surface of the body is: Mucous membrane

- ~~A.~~ Cutaneous membrane
- ~~B.~~ Meningeal membrane
- ☒ C. Serous membrane

- D. Mucus membrane
- E. All of the above

28. The membrane of which organelle is continuous with the outer nuclear membrane?

- A. Centrioles
- B. Microtubule
- C. Golgi apparatus
- ☒ D. Rough endoplasmic reticulum
- E. Smooth endoplasmic reticulum

29. Vesicles produced from the rough endoplasmic reticulum will be transported to which organelle?

- A. Smooth endoplasmic reticulum
- B. Lysosome
- C. Peroxisome

- ☒ D. Golgi apparatus
- E. Microtubule

30. Which of the following is not a function of smooth endoplasmic reticulum?

- A. Synthesis of steroid hormones ✓
- B. Detoxification ✓
- C. Store and release calcium ions in the striated muscle cells ✓
- D. All of the above
- ☒ E. None of the above

31. Which of the following is **wrong** about the microtubules?

19. A unique fibrous type of joint between a tooth & its socket
- A. Suture
 - B. Syndesmoses
 - C. Gomphoses
 - D. Symphyses
 - E. None of the above
20. Which one of the following is not a factor influencing synovial joint stability?
- A. The nature of the articular surfaces ✕
 - B. The number and positioning of the ligaments ✕
 - C. The tone and strength of the muscles acting upon the joint
 - D. The type of joint
 - E. None of the above
21. A type of synovial joint with a cylindrical shaped projection of one bone fits into a trough shaped surface of another bone with movement occurs within a single plane
- A. Plane
 - B. Hinge
 - C. Condylloid ✕
 - D. Saddle ✕
 - E. None of the above
22. Which one describes the condition in which an independent bones that fuse with other bones during development and remains as part of that bone?
- A. Pressure epiphysis
 - B. Traction epiphysis
 - C. Apophysis
 - D. Atavistic epiphysis
 - E. None of the above
23. The following are connective tissue layer around the skeletal muscle. Identify the correct order from internal to external
- A. Endomysium - perimysium - epimysium
 - B. Epimysium - perimysium - endomysium
 - C. Perimysium - endomysium - epimysium
 - D. Endomysium - epimysium - perimysium
 - E. None of the above
24. The ability of muscle tissue to return to original shape and length after contraction and stretching is known as

- C. More than one chromosomes are found in one prokaryotic cells
- ☒ D. Organelles are found in the different compartments in eukaryotic cells
- E. The outer and the inner membranes of eukaryotic cells are identical

13. Which of the following is true about the integral proteins in the cell membrane

- A. Easily removed without destroying the cell membrane
- ☒ B. Serve as ion channels and receptors
- C. Found in the inner membrane facing towards the cytoplasm
- D. Not cross the whole cell membrane
- E. Not penetrate the hydrophobic core of the membrane

14. Which of the following is not the function of the membrane proteins

- A. Signal transduction ✓
- B. Communication ✓
- C. Recognition ✓
- D. Transportation ✓
- ☒ E. None of the above

15. Why nucleus controls the cell activities

- A. Found at the center of the cytoplasm ✓
- B. Being the largest organelle ✗
- ☒ C. Contains genetic material ✓
- D. Having double membrane ✗
- E. Having nuclear envelop ✗

16. The plane that separates the body into unequal right and left sides is

- A. Transverse plane
- B. Midsagittal plane
- ☒ C. Parasagittal plane
- D. Frontal plane
- E. Coronal plane

17. The only bone that does not articulate with any other bone

- A. Sternum
- B. Nasal bone
- C. Vomer
- ☒ D. Hyoid bone
- E. None of the above

18. Which one of the following bone is not part of appendicular skeleton?

- A. Scapula ✓
- B. Hip bone ✓
- C. Clavicle ✓
- D. Humerus ✓
- ☒ E. Sacrum

INSTRUCTION 1: READ THE FOLLOWING QUESTIONS AND CHOOSE THE BEST ANSWER AND WRITE THE LETTER OF YOUR CHOICE ON THE SEPARATE ANSWER SHEET PROVIDED.

- Which one of the following statement is not correct concerning circulatory system?
 - Capillaries are always located at the end of arterioles & beginning of venules
 - Exchange of substances with extracellular fluid occurs at the capillary bed ✓
 - Arteries in the retina are true terminal arteries ✓
 - Pulmonary veins are atypical veins
 - Lymphatic plexuses are absent in some tissues of the body
- Which of the following show the appropriate respective section thicknesses for light and electron microscope?

☒ A. 5 – 10 μm and 40 – 70 nm	D. 100 – 200 nm and 25 – 50 μm
B. 40 – 70 nm and 5 – 10 μm	E. e) 25 – 50 mm and 100 – 200 μm
C. 25 – 50 μm and 100 – 200 nm	
- Which of the following is a commonly used clearing agent in tissue section preparation for light microscopy?

X A. Ethanol	D. Formalin	fixation - for dehy - alc clear -
B. Methanol	E Xylen	
C. Paraffin wax		
- In hematoxylin and eosin method of the general tissue section staining, hematoxylin:

A. Acts like an acidic dye	C. Stains blue black
B. Stains best the ribosomes	D. a & b
	E b & c
- Which one of the following is the correct order of tissue block preparation from an autopsy tissue?
 - Dehydration, fixation, clearing, infiltration & embedding
 - Dehydration, clearing, infiltration, fixation, & embedding
 - ☒ Fixation, dehydration, clearing, infiltration & embedding
 - Fixation, dehydration, infiltration, clearing & embedding
 - Clearing, dehydration, fixation, infiltration & embedding
- Among the different lenses in a bright field microscope, which one functions in resolution of a tissue image?

Module I part I final exam answer sheet

Name Which one is not correct about Circulatory ID NO. _____

April 24, 2015

I. Multiple choice

II. Matching

1	A	23	A	45	C	67	D	1	F
2	A	24	D	46	A	68	A	2	E
3	E	25	C	47	A	69	E	3	B
4	E	26	B	48	D	70	D	4	D
5	C	27	C Holtz	49	B	71	D	5	A
6	A	28	D	50	A	72	B		
7	B	29	D	51	D	73			
8	A	30	E	52	D				
9	E	31	C	53	B				
10	C	32	D	54	C				
11	D	33	D	55	C				
12	D	34	C	56	B				
13	B	35	A	57	C				
14	E	36	D	58	D				
15	C	37	B	59	C				
16	C	38	B	60	B				
17	D	39	B	61	A				
18	E	40	A	62	B				
19	C	41	C	63	A or D				
20	E	42	E	64	E				
21	B	43	A	65	—				
22	D	44	C	66	C				